

NUMA2.pdf



abcd__1234



Paralelismo



3º Grado en Ingeniería Informática



Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya



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 **Gemini 3**
Nuestro modelo más preciso

Sintetiza horas de
Investigación en minutos

Necesito estudiar a fondo el comportamiento de la fotosíntesis según el tipo de planta y el entorno



 Deep Research 



 Fuentes 

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jueves, 26 de
febrero de 2026,
16:00 - 19:00

Pregunta 1

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

Given a NUMA multiprocessor system with 4 nodes, each node with one processor, 64 KB of private cache and 8 GB of main memory. Coherence among nodes is kept with a directory-based MSU protocol, using the transactions explained in class.

Assume that the home node for variable *var* is NUMAnode0 and that at a given moment there exist clean copies of that variable in the cache memories in NUMAnode1 and NUMAnode2, but no copies in NUMAnode0 and NUMAnode3. If the processor in NUMAnode1 wants to update (write to) variable *var*, which of the following coherence actions do NOT occur?

Trieu-ne una o més:

- ☐ The processor in the local node (NUMAnode1) issues PrWr.
- ☐ The local node sends a UpgrReq command to the home node (NUMAnode0).
- ☐ Since there is another copy of the line in the cache of a remote node (NUMAnode2), the home node sends an Invalidate command to it.
- ☒ Since there is another copy of the line in the cache of a remote node (NUMAnode2), the local node sends an Invalidate command to it.

Pregunta 2

Completa

Puntuació 1,00
sobre 1,00

Marca la
pregunta

After the previous access, the processor in NUMAnode3 reads the same variable *var*. Which of the following coherence actions do NOT occur?

Trieu-ne una o més:

- ☐ The processor in the local node (NUMAnode3) issues PrRd.
- ☒ The local node issues a Fetch command to the home node (NUMAnode0) to obtain the line containing *var*.
- ☐ The home node then issues a Fetch command to the remote NUMAnode1 in order to get the most up-to-date copy of the requested memory line.
- ☐ The remote node issues a Dreply command with the only valid copy of the line in the whole system.

WUOLAH